

The integration mechanism of the four positional cues—anterior, posterior, dorsal, and ventral—in axolotl limb regeneration

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eLife Assessment

This **fundamental** work by Yamamoto and colleagues advances our understanding of how positional information is coordinated between axes during limb outgrowth and patterning. They provide **solid** evidence that the dorsal-ventral axis feeds into anterior-posterior signaling, and identify the responsible molecules by combining transplantations with molecular manipulations. This work will be of broad interest to regeneration, tissue engineering, and evolutionary biologists.

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Abstract

Axolotls (*Ambystoma mexicanum*) exhibit a remarkable ability to regenerate limbs after amputation. Classical experiments have suggested that the integration of four positional cues—dorsal, ventral, anterior, and posterior—within a regenerating blastema is necessary for accurate limb pattern formation. However, the molecular basis underlying this integration has remained largely unknown. Here, we demonstrate that both dorsal and ventral tissues are required for limb formation via induction of *Shh* expression, which plays a crucial role in limb patterning. Using the accessory limb model (ALM), we induced position-specific blastemas lacking cells derived from a single orientation (anterior, posterior, dorsal, or ventral). Patterned limb formation occurred only in blastemas containing both dorsal- and ventral-derived cells. We further observed that *Shh* expression requires dorsoventral contact within a blastema, highlighting the necessity of dorsoventral contact for inducing *Shh* expression. Additionally, we identified molecules mediating dorsoventral-dependent *Shh* expression: WNT10B as the dorsal factor and FGF2 as the ventral factor. Our findings clarify the role of dorsal and ventral positional cues in inducing *Shh*, a mechanism that has rarely been studied in the context of limb regeneration and pattern formation. This model provides new insights into how interpositional interactions are integrated to drive the regeneration process.

Introduction

Axolotls (*Ambystoma mexicanum*) possess remarkable regenerative abilities, enabling the regeneration of entire limbs after amputation. Among tetrapods (four-limbed vertebrates), only urodele amphibians retain the lifelong ability to regenerate fully developed limbs. Investigating the molecular mechanisms underlying axolotl limb regeneration could provide valuable insights for advancing regenerative medicine in humans, potentially leading to new therapies for tissue repair and organ regeneration.

The limb regeneration process can be divided into two stages: the blastema (regenerating limb primordium) induction process and the limb patterning process. The induction of a blastema, which contains highly proliferative multipotent and unipotent cells, depends on the presence of nerves at the injured region. When a denervated limb is amputated, a blastema is not induced (Todd, 1823 [\[1\]](#)). The blastema induction process is followed by the limb patterning process. In limb development, positional values are thought to be established along three-dimensional axes—anteroposterior, dorsoventral, and proximodistal—based on positional information. For example, WNT7A, secreted from dorsal ectoderm, acts as dorsal positional information to induce *Lmx1b* expression, which acts as dorsal positional value, in underlying mesenchymal cells (Riddle et al., 1995 [\[2\]](#); Vogel et al., 1995 [\[3\]](#); Cygan et al., 1997 [\[4\]](#); Chen et al., 1998 [\[5\]](#); Chen, 2002 [\[6\]](#)). Similarly, SHH is known to serve as posterior positional information, regulating anteroposterior patterning (Riddle et al., 1993 [\[7\]](#)). For successful limb patterning in limb regeneration, interactions among anterior, posterior, dorsal, and ventral cells within a blastema have been considered essential. The importance of these interpositional interactions was investigated in creating double-half limbs. Amputating the double-half limbs can provide a blastema in a “one-positional-value-missing” state. For example, when a double-dorsal limb is amputated, the resulting blastema lacks ventral positional value. Such blastemas form hypomorphic, spike-like structures or fail to regenerate entirely (Bryant, 1976; Bryant and Baca, 1978 [\[8\]](#); Burton et al., 1986 [\[9\]](#)). It is noteworthy that structural patterns along the anteroposterior axis have been lost, even though both anterior and posterior cells should be present in a blastema induced from double-dorsal and double-ventral limb. Additional experiments involving ectopic contact between opposite regions (e.g., anterior-posterior or dorsal-ventral) of a blastema further support the critical role of interpositional interactions in the limb patterning process, as such interactions can lead to ectopic limb formation (Iten and Bryant, 1975 [\[10\]](#); Bryant and Iten, 1976 [\[11\]](#); Maden, 1980 [\[12\]](#); Stocum, 1982 [\[13\]](#)). These findings suggest that specific signals from all four positional domains must be integrated for successful limb patterning, such that the absence of any one of them leads to failure. In the present study, we define these position-specific signals required for a blastema to form a limb as “positional cues.” We distinguish “positional cues” from “positional information” to clarify their distinct roles: positional information refers to signals that induce positional values within recipient cells, whereas positional cues do not necessarily induce positional values and may act downstream of positional values. For example, WNT7A from the dorsal ectoderm in amniote limb development functions as dorsal positional information because it induces the dorsal positional values within the underlying mesenchyme. In contrast, FGF8 from the anterior region in axolotl limb regeneration functions as the anterior positional cue because it can substitute for anterior tissues in the limb patterning process (Nacu et al., 2016 [\[14\]](#)). Of course, these definitions are not mutually exclusive. One important feature of positional cue molecules is that they can substitute for tissues of one orientation in limb regeneration. To further understand limb regeneration, the molecular basis of the integration of the positional cues in limb patterning process should be elucidated.

The accessory limb model (ALM), a non-amputation experimental system, provides a valuable approach for studying axolotl limb regeneration, particularly the role of interpositional interactions (Endo et al., 2015 [\[15\]](#)). In this model, a blastema is induced through skin wounding

combined with nerve deviation (Endo et al., 2004 [↗](#), Satoh et al., 2007 [↗](#)). When ALM surgery is performed on the anterior side, the induced blastema (ALM blastema) lacks posterior-derived cells and is unable to form a properly patterned limb. This limitation can be overcome by supplying posterior-derived cells via a skin graft from the posterior side, enabling the successful regeneration of a patterned limb. Similarly, an ALM blastema can be induced in a position-specific manner along the limb axes. In this case, the induced ALM blastema will lack cells from the opposite side. This condition allows the investigation of blastema induction and limb patterning by selectively removing cells from specific orientations of the limb. Analyzing the characteristics and developmental limitations of these ALM blastemas provides valuable insights into the positional cues that regulate axolotl limb regeneration.

The molecular basis of the interpositional interactions within a blastema in the limb patterning process has been partially elucidated (Nacu et al., 2016 [↗](#)). In a normal blastema, *Fgf8* and *Shh* are expressed at the anterior and the posterior regions, respectively. A previous study demonstrated that FGF8 and SHH proteins can substitute for anterior and posterior tissues, respectively, to form a patterned limb. Moreover, these proteins function as mutually inductive signals, maintaining each other's expression. These findings suggest that FGF8 and SHH serve as the anterior and posterior positional cues, respectively. In contrast, the molecular basis of dorsal and ventral positional cues remains poorly understood. Previous studies have shown that retinoic acid supplementation can convert the dorsal positional value to the ventral, enabling patterned limb formation even in the absence of ventral tissues (Vieira et al., 2019 [↗](#)). However, the ventral positional cue remains uncertain, as retinoic acid induces ventral positional values rather than acting as the ventral positional cue itself, suggesting the existence of another, downstream ventral positional cue. Consequently, the mechanisms underlying dorsal and ventral positional cues, as well as their integration with anterior and posterior cues, remain unclear.

In the present study, we investigated the roles of dorsal and ventral positional cues in limb patterning. We confirmed the necessity of dorsoventral tissue contact for limb patterning using the ALM. We found that the induction of *Shh* expression was dependent on the co-existence of dorsal and ventral cells in axolotl limb regeneration. Furthermore, we identified *Wnt10b* and *Fgf2* as dorsal and ventral positional cues, respectively, by RNA-seq analysis, and confirmed that these genes mediate *Shh* expression in axolotl blastemas. Our results suggest that the crucial role of the dorsal and ventral positional cues in the limb patterning process is to induce *Shh* expression, thereby enabling anteroposterior interaction. This model provides a framework for understanding the integration of the four positional cues during axolotl limb regeneration.

Results

Dorsal-ventral tissue contact is required for limb regeneration in the ALM

We performed the ALM experiment on *Ambystoma mexicanum* to investigate whether dorsoventral tissue contact, as well as anteroposterior tissue contact, is required for limb regeneration (Fig. 1 [↗](#)). The skin was removed from one side (anterior, posterior, dorsal, or ventral) of a limb, and the large nerve fibers running along the center of a limb (*Nervus medianus* and *N. ulnaris*, Fig. 1A [↗](#)) were dissected and rerouted to the wounded region. As mentioned before, we supplied the cells from the contralateral side of a limb as a skin graft in experimental conditions to ensure the presence of sufficient cells for patterned limb formation. We conducted eight types of ALM blastema inductions, which can be categorized into two groups: basic ALM blastemas and skin-grafted ALM blastemas. The basic ALM blastemas included anteriorly induced ALM blastema (AntBL), posteriorly induced ALM blastema (PostBL), dorsally induced ALM blastema (DorBL), and ventrally induced ALM blastema (VentBL), all of which should lack cells from the contralateral side. The skin-grafted ALM blastemas included AntBL with posterior skin

grafting (AntBL+P), PostBL with anterior skin grafting (PostBL+A), DorBL with ventral skin grafting (DorBL+V), and VentBL with dorsal skin grafting (VentBL+D); these ALM blastemas should contain cells with all anteroposterior and dorsoventral orientations. For the ALM experiment, we defined the two large blood vessels running on the anterior and posterior sides at the stylopod level (*Vena cephalica* and *V. basilica*, **Fig. 1A**) as the anterior and posterior dorsoventral borders, respectively. At 10 days post-surgery (dps), an ALM blastema was observed in all experimental groups (**Fig. 1B–I**). Consistent with previous studies (Endo et al., 2004; Nacu et al., 2016), limbs with multiple digits were formed from AntBL+P and PostBL+A (**Table 1**, **Fig. 1F, G**), whereas AntBL and PostBL either regressed or formed bump structures (**Table 1**, **Fig. 1B, C**). Similarly, DorBL+V and VentBL+D formed limbs (**Table 1**, **Fig. 1H, I**), whereas most DorBL and VentBL either regressed or formed bump structures (**Table 1**, **Fig. 1D, E**). Notably, only 1 out of 12 DorBL formed a limb, which may have resulted from ventral tissue contamination, as the rerouted nerves originally ran along the ventral side of the humerus. In contrast, none of 22 VentBL formed multi-digit limbs. These results suggest that both dorsal and ventral tissues, as well as anterior and posterior tissues, are required for successful limb formation.

Gene expression patterns in ALM blastemas

To elucidate the characteristics of the ALM blastemas that typically fail to form limbs (AntBL, PostBL, DorBL, and VentBL), we analyzed gene expression patterns at 10 dps (**Fig. 2**). *Lmx1b*, a gene necessary and sufficient for establishing the dorsal identity in mesenchymal cells in developing limb buds (Vogel et al., 1995; Chen et al., 1998; Chen, 2002), was used as a dorsal marker because we previously reported that *Lmx1b* is exclusively expressed in dorsal-derived cells and can be activated in the ALM experiment (Iwata et al., 2020; Yamamoto et al., 2022). The expression patterns of *Fgf8*, *Shh*, and *Lmx1b* were investigated using *in situ* hybridization (ISH). In AntBL and PostBL, *Lmx1b* expression was restricted to the dorsal half of the blastemas (**Fig. 2B, G**; $n=5/5$ for both), suggesting the presence of both dorsal and ventral tissues. These expression patterns correspond to the anatomically defined dorsoventral borders (**Fig. 1A**). In contrast, *Lmx1b* was expressed throughout the entire region of DorBL (**Fig. 2L**, $n=6/6$), whereas no *Lmx1b* expression was detected in VentBL (**Fig. 2Q**, $n=6/6$), suggesting the absence of ventral tissue in DorBL and dorsal tissue in VentBL. Consistent with previous studies, *Fgf8* expression was observed in AntBL (**Fig. 2C**, $n=4/5$), but not in PostBL (**Fig. 2H**, $n=5/5$; Nacu et al., 2016). Conversely, *Shh* expression was detected in PostBL (**Fig. 2I**, $n=5/5$), but not in AntBL (**Fig. 2D**, $n=5/5$). In DorBL and VentBL, *Shh* expression was largely absent (**Fig. 2N**, $n=5/6$; **Fig. 2S**, $n=6/6$), whereas *Fgf8* expression was present in both (**Fig. 2M**, $n=4/6$; **Fig. 2R**, $n=4/6$). In DorBL, *Shh* expression was observed in only 1 out of 6 samples, possibly due to ventral tissue contamination, as described above. These results suggest that *Fgf8* expression is independent of the co-existence of dorsal and ventral tissues, whereas *Shh* expression appears to require their presence.

Induction of *Shh* expression requires both dorsal and ventral cells

The absence of *Shh* expression in most DorBL and VentBL, combined with its consistent expression in all PostBL—which contain both *Lmx1b*-positive and *Lmx1b*-negative regions—suggests that the induction of *Shh* expression depends on the presence of both dorsal and ventral cells (**Fig. 2**). To test this hypothesis, we performed cell-tracing experiments using GFP-expressing skin from transgenic animals (**Fig. 3**). We induced VentBL in wild-type animals and then grafted the posterior half of the dorsal skin (VentBL+PD_{gfp}) or posterior half of the ventral skin (VentBL+PV_{gfp}) from GFP transgenic animals (**Fig. 3A**). Similarly, DorBL was induced in wild-type animals, and the posterior half of the dorsal skin (DorBL+PD_{gfp}) or the posterior half of the ventral skin (DorBL+PV_{gfp}) from GFP transgenic animals was grafted (**Fig. 3A**). GFP-positive mesenchymal cells derived from the posterior skin were expected to express *Shh* if provided with a suitable environment, because posteriorly derived cells are known to have the competency to express *Shh* in a blastema (Iwata et al., 2020). If the induction of *Shh* expression depends on the co-existence

Figure 1.

ALM experiments at the four positions.

(A) Schematic image of anatomy at the stylopod level of axolotl limb. HE staining (bright field) and acetylated alpha tubulin, visualized by immunofluorescence (green, dark field), are shown in the right panels. Black and white arrows, respectively, indicate major blood vessels and nerves. H: *humerus*, AHL: *Anconaeus humeralis lateralis*, HAB: *Humeroantebrachialis*, CBL: *coracobrachialis longus*. (B–I) Blastemas induced at the anterior, posterior, dorsal, or ventral region by skin wounding plus nerve deviation without (B–E) or with (F–I) skin grafting from the opposite side of the limb. Images were captured at 10 and 60 dps. Scale bar = 3 mm.

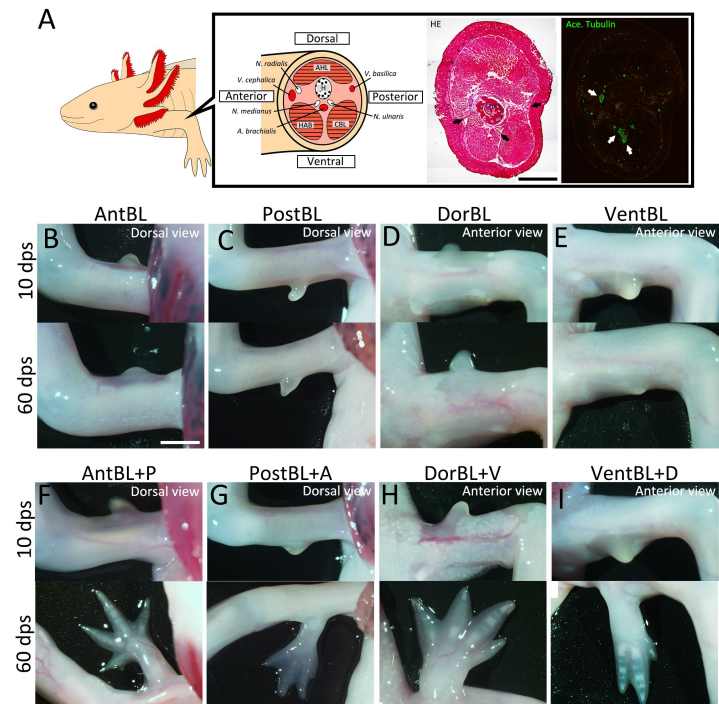


Table 1.

The induction rate of bump/limb formation in ALM.

	<i>N</i>	Regress		Bump		Limb	
AntBL	8	3	(37.5 %)	5	(62.5 %)	0	(0.0 %)
AntBL+P	9	0	(0.0 %)	1	(11.1 %)	8	(88.9 %)
PostBL	8	6	(75.0 %)	2	(25.0 %)	0	(0.0 %)
PostBL+A	7	2	(28.6 %)	1	(14.3 %)	4	(57.1 %)
DorBL	12	8	(66.7 %)	3	(25.0 %)	1	(8.3 %)
DorBL+V	14	5	(35.7 %)	2	(14.3 %)	7	(50.0 %)
VentBL	22	15	(68.2 %)	7	(31.8 %)	0	(0.0 %)
VentBL+D	11	2	(18.2 %)	2	(18.2 %)	7	(63.6 %)

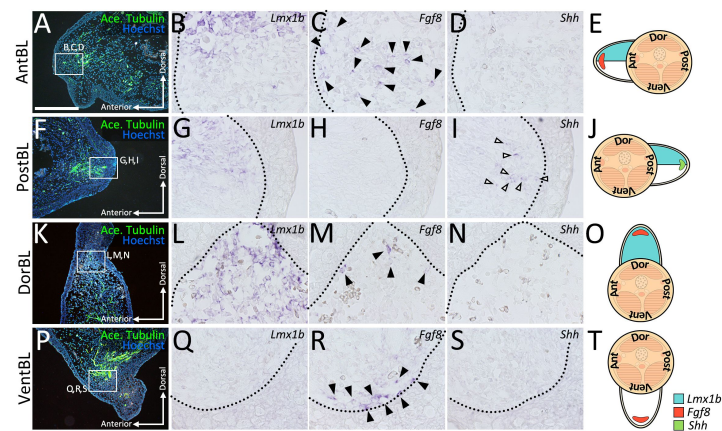


Figure 2.

Gene expression patterns of the ALM-induced blastemas.

Sections of anteriorly (A–D), posteriorly (F–I), dorsally (K–N), or ventrally (P–S) induced blastemas at 10 dps. Acetylated alpha tubulin (A–P) was visualized by immunofluorescence. Expression of *Lmx1b* (B–Q), *Fgf8* (C–R), and *Shh* (D–S) in the regions indicated by white boxes in (A–P) was visualized by *in situ* hybridization. Black and white arrowheads indicate the signals of *Fgf8* and *Shh* expression, respectively. The dotted line indicates the epithelial-mesenchyme border. (E–T) Schematic images of gene expression patterns. Scale bar = 700 μm.

of dorsal and ventral cells, *Shh* expression in the GFP-positive cells should be observed in VentBL+PD_{gfp} and DorBL+PV_{gfp} but not in DorBL+PD_{gfp} or VentBL+PV_{gfp}. Samples were collected at 10 dps, and the expression of *Shh* and *Lmx1b* was analyzed using ISH. In all four experimental groups, *Lmx1b* expression patterns in the GFP-negative regions were consistent with those in DorBL and VentBL (Fig. 3D, F, H, J). In DorBL+PD_{gfp} and VentBL+PD_{gfp}, *Lmx1b* expression was observed in GFP-positive mesenchymal cells, whereas GFP-positive cells in DorBL+PV_{gfp} and VentBL+PV_{gfp} were *Lmx1b*-negative (Fig. 3C–F). These *Lmx1b* expression patterns suggest that dorsoventral tissue contact was present in VentBL+PD_{gfp} and DorBL+PV_{gfp} but absent in DorBL+PD_{gfp} and VentBL+PV_{gfp}. In the GFP-positive cells derived from PD_{gfp} skin, *Shh* expression was observed in VentBL+PD_{gfp} (Fig. 3C, n=3/3) but not in DorBL+PD_{gfp} (Fig. 3E, n=3/3). Similarly, in the GFP-positive cells derived from PV_{gfp} skin, *Shh* expression was observed in DorBL+PV_{gfp} (Fig. 3I, n=3/6) but not in VentBL+PV_{gfp} (Fig. 3G, n=7/7). In addition, *Shh* expression was observed in some GFP-negative mesenchymal cells in samples where *Shh* expression was observed in GFP-positive cells. These results suggest that the induction of *Shh* expression in posteriorly derived cells requires the co-existence of dorsal and ventral cells.

Limb formation in the absence of dorsoventral tissue contact

DorBL and VentBL failed to form limbs (Fig. 1D, E). *Shh* expression was absent in these blastemas, whereas *Fgf8* was expressed (Fig. 2). Results from the cell-tracing experiments suggest that *Shh* expression requires the co-existence of dorsal and ventral cells (Fig. 3). Based on these findings, we hypothesized that while the co-existence of dorsal and ventral cells is necessary to induce *Shh* expression, it is not directly required for limb patterning if SHH proteins are externally provided. To test this hypothesis, we overexpressed *Shh* in DorBL and VentBL (Fig. 4A, B). As a positive control, *Shh* was overexpressed in AntBL, where *Fgf8* was expressed but *Shh* expression was absent (Fig. 2C, D), as a previous study has shown that SHH supplementation to AntBL can induce limb patterning (Nacu et al., 2016). As a result, *Shh*-electroporated AntBL, DorBL, and VentBL formed limbs (n=10/19, 6/18, and 8/14, respectively, Fig. 4C–G). In contrast, GFP-electroporated DorBL and VentBL (negative controls) failed to form limbs (n=7/7 and 6/6, respectively). We further analyzed the anatomy of the induced limbs (Fig. 4H–K). We found that limbs formed from DorBL and VentBL exhibited symmetric structures along the dorsoventral axis, suggesting double-dorsal and double-ventral structures, respectively (Fig. 4J, K). In contrast, limbs formed from AntBL, which contained both *Lmx1b*-positive and *Lmx1b*-negative regions (Fig. 2B), exhibited asymmetric structures similar to those of normal limbs (Fig. 4H, I). To evaluate the symmetry of the limbs, we applied a machine learning-based method using ilastic. Each pixel in the images was classified into five classes (Classes 1 to 5, Fig. 4L). In this process, we annotated regions of background (Class 1), cartilage (Class 2), muscle (Class 3), other connective tissue (Class 4), and epidermis (Class 5) as training data for classification. Using these annotated regions as references, pixels were automatically classified into the five classes. Each class was assumed to primarily represent these tissues and regions with similar characteristics. Symmetry scores were then calculated for each class individually and for the combined set of all classes (Fig. 4M–R, see Materials and Methods). The analysis revealed that symmetry scores for Classes 2 and 3, which encompass cartilage and muscle, and scores for the combined set of all classes were significantly higher in limbs formed from DorBL and VentBL compared to intact limbs (Fig. 4N, O, R). In contrast, the differences in symmetry scores for Classes 1, 4, and 5 in these limbs were relatively small (Fig. 4M, P, Q), likely because the axis of symmetry was manually set to maximize the external shape as symmetrically as possible. No significant differences in symmetry scores across all classes were observed between intact limbs and limbs formed from AntBL. These results suggest that limbs formed from DorBL and VentBL exhibit symmetric internal structures compared to normal limbs. Therefore, SHH supplementation appears to compensate for the lack of co-existence of dorsal and ventral cells in limb patterning, without inducing new dorsal or ventral identities. Thus, we conclude that the co-existence of both dorsal and ventral cells is critical for inducing *Shh* expression, which in turn is essential for limb patterning.

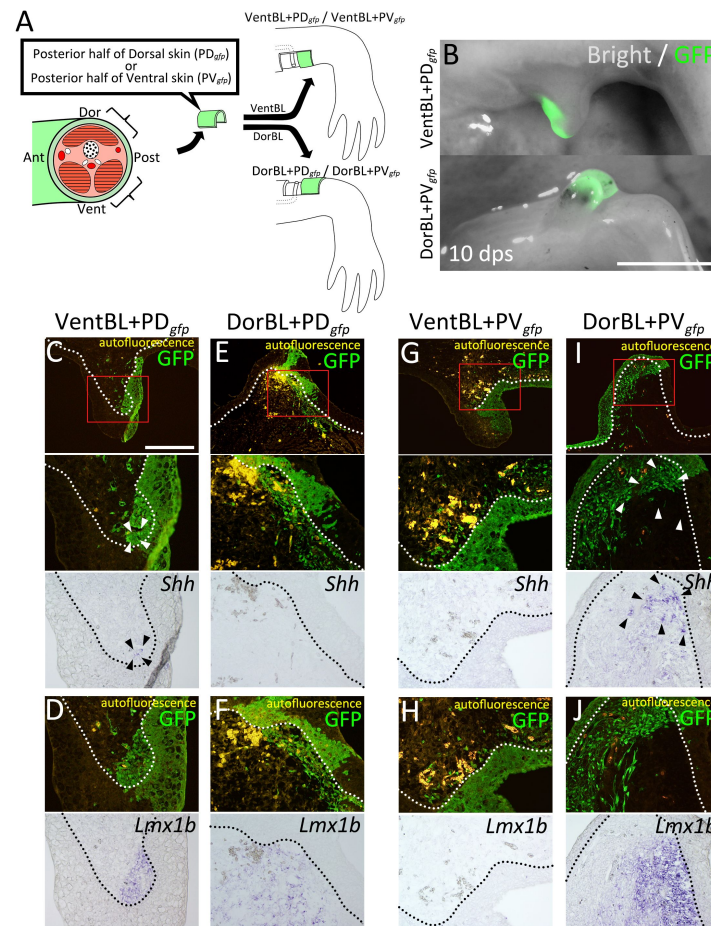


Figure 3.

Co-existence of dorsal and ventral cells induces *Shh* expression.

(A) Experimental scheme. Posterior half of dorsal (PD_{gfp}) or ventral (PV_{gfp}) GFP-expressing skin was grafted on VentBL (VentBL+PD_{gfp}; C, D / VentBL+PV_{gfp}; G, H), or DorBL (DorBL+PD_{gfp}; E, F / DorBL+PV_{gfp}; I, J) region. (B) Induced blastemas at 10 dps; images of bright and dark fields are merged. (C–J) Dark and bright fields of the same sections of induced blastemas at 10 dps. Red boxes in (C–I) indicate the corresponding regions of lower images. Expressions of *Shh* and *Lmx1b* were visualized by *in situ* hybridization. GFP signals were visualized by immunofluorescence. Arrowheads indicate the cells expressing *Shh*. The dotted line indicates the epithelial-mesenchyme border. Scale bar = 3 mm (B) and 700 μm (C).

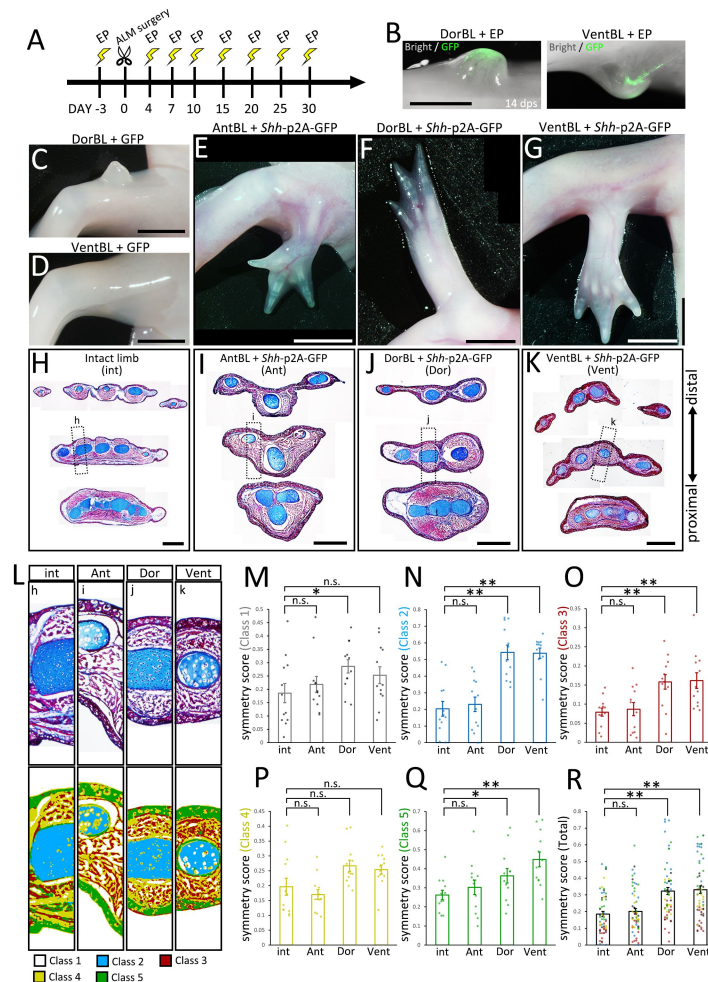


Figure 4.

Limb formation without the co-existence of dorsal and ventral cells by SHH supplementation.

(A) Experimental scheme. *Shh*-p2A-GFP- or GFP-containing pCS2 vector was electroporated (EP) into AntBL, DorBL, and VentBL. (B) Induced blastemas at 14 dps; images of bright and dark fields are merged. (C–G) Phenotypes at 90 dps. (H–K) Histological analysis of intact and induced limbs. Standard Masson's trichrome staining was performed on the transverse sections. The dotted boxes indicate the regions shown in (L). (L) Upper panels: analyzed regions for calculating symmetry scores. Lower panels: images after pixel classification by machine learning. (C–I) Symmetry scores of each class. Scores obtained from the same limb are plotted at the same x-coordinates. n.s.; no significant difference, * $p < 0.05$, ** $p < 0.005$. Scale bar = 2 mm (B), 4 mm (C–G), and 1 mm (H–K).

The molecular basis of the dorsal and ventral positional cues

To investigate the molecular basis of dorsal and ventral positional cues, we performed RNA-seq analysis on DorBL and VentBL and identified differentially expressed genes (DEGs) between the two groups ($p < 0.05$, **Fig. 5A**). Among the DEGs, we specifically focused on genes annotated as “intercellular signaling molecules” in PANTHER and identified 21 genes (**Fig. 5B**). In this analysis, we found that 5 genes were expressed at higher levels in DorBL and 16 genes were expressed at higher levels in VentBL (**Fig. 5B**). Notably, *Wnt4*, *Wnt10b*, *Fgf2*, *Fgf7*, and *Tgfb2*, which encode secreted proteins, belong to the WNT, FGF, and TGFB families, which play key roles in major signaling pathways regulating limb developmental processes. To examine whether these genes regulate the induction of *Shh* expression in ALM blastemas, we overexpressed each gene in either DorBL or VentBL. As a result, *Shh* expression was observed in *Wnt10b*-electroporated VentBL ($n=3/3$) and *Fgf2*-electroporated DorBL ($n=5/7$, **Fig. 5C**). In contrast, *Shh* expression was not detected in *Wnt10b*-electroporated DorBL ($n=4/4$) or *Fgf2*-electroporated VentBL ($n=3/3$). Similarly, *Shh* expression was not detected in *Fgf7*- or *Tgfb2*-electroporated DorBL ($n=3/3$ for both) or in *Wnt4*-electroporated VentBL ($n=4/4$). To confirm *Shh* induction by FGF2 in DorBL, we supplied commercially obtained mouse FGF2 proteins into DorBL via bead grafting and observed higher *Shh* expression in DorBL with FGF2-soaked bead compared to DorBL with control DDW-soaked bead (**Fig. 5D**). Next, to confirm *Shh* induction by WNT signaling in VentBL, we treated VentBL with 1 μ M 6-bromoindirubin-3-oxime (GSK-3 Inhibitor, BIO). As a result, *Shh* expression was relatively higher in BIO-treated VentBL compared to DMSO-treated control VentBL (**Fig. 5E**). Additionally, symmetric limbs were formed in *Wnt10b*-electroporated VentBL ($n=3/12$), BIO-treated VentBL ($n=3/8$), and in *Fgf2*-electroporated DorBL ($n=5/10$), being consistent with the results of *Shh* overexpression (**Fig. 5F–H**, **Fig. S1**). These findings suggest that WNT10B, expressed highly in dorsal blastema cells, and FGF2, expressed highly in ventral blastema cells, function as the dorsal and ventral positional cues, respectively, to induce *Shh* expression, which subsequently supports limb patterning.

Limb formation without nerve deviation and ventral skin grafting

A previous study demonstrated that a cocktail of BMP2, FGF2, and FGF8 can substitute for nerve deviation in the ALM experiment on the anterior region, suggesting that nerves contribute to blastema induction by supplying these proteins (Makanae et al., 2014). We found that FGF2 can also substitute for ventral tissue in DorBL (**Fig. 5**), suggesting that FGF2 serves not only as part of the nerve factors in blastema induction but also as the ventral positional cue in limb patterning. To investigate whether supplementation with BMP2, FGF2, and FGF8 to the dorsal region could substitute for both nerve deviation and ventral skin grafting, we performed experiments involving gelatin beads soaked in these proteins (**Fig. 6**). The dorsal or ventral skin was removed at the zeugopod level, and a BMP2+FGF2+FGF8-soaked gelatin bead was grafted at 3 dps without nerve deviation. Blastema induction was observed in both experimental groups (**Fig. 6A, B**, upper panels). We investigated gene expression patterns in the induced blastemas and found that both *Fgf8* and *Shh* were expressed in blastemas induced at the dorsal site ($n=8/8$), whereas *Shh* expression was absent in ventral blastemas ($n=5/5$), although *Fgf8* expression was detected in most cases ($n=4/5$, **Fig. 6C, D, F, G**). *Lmx1b* expression patterns corresponded to those observed in DorBL and VentBL (**Fig. 6E, H**, $n=8/8$ and $5/5$, respectively). Furthermore, limb patterning was observed in the dorsal groups ($n=12/20$) but not in the ventral groups ($n=17/17$, **Fig. 6A, B**, lower panels). These findings demonstrate that a straightforward procedure involving dorsal skin wounding and supplementation with BMP2, FGF2, and FGF8 is sufficient to induce accessory limb formation. Remarkably, this method also enabled the induction of multiple limbs within the same limb. Grafting BMP2+FGF2+FGF8-soaked beads to multiple dorsal sites resulted in the formation of multiple limbs along the proximodistal axis (**Fig. 6I–L**). These results refine our understanding of the role of FGF2 in the ALM, particularly its critical function on the dorsal side.

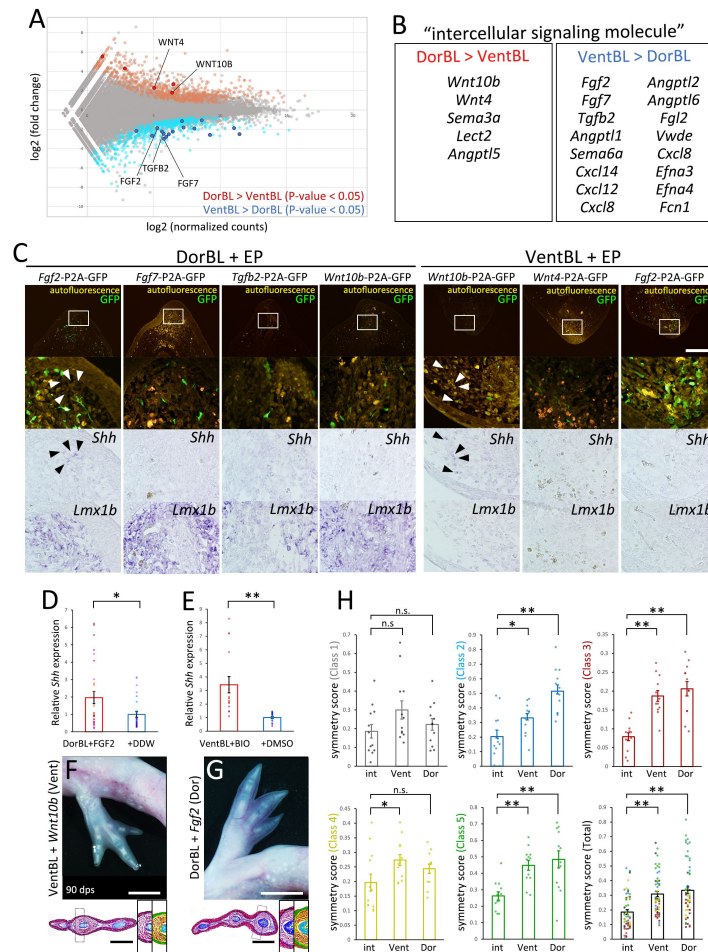


Figure 5.

Identification of candidate molecules of the dorsal and ventral positional cues.

RNA-Seq was performed on DorBL and VentBL at 10 dps. (A) MA plot of the result. (B) List of DEGs annotated as “intercellular signaling molecules.” (C) Bright and dark fields of sections of DorBL and VentBL at 10 dps with candidate genes introduced. Expression of *Shh* and *Lmx1b* was visualized by *in situ* hybridization, and GFP signals were visualized by immunofluorescence. Arrowheads indicate the cells expressing *Shh*. The white boxes indicate the regions of the lower panels. Images of dark and bright fields of *Shh* are obtained from the same section. (D, E) Quantitative analysis of *Shh* expression in DorBL with FGF2-soaked bead and with DDW-soaked bead (D) and in VentBL treated with 1μM BIO and with DMSO (E). RNA was prepared from 8 identical DorBL for both (D) and from 4 identical VentBL for both (E). Technological replicates are plotted in the same color. (F, G) The limbs formed from VentBL with *Wnt10b* and DorBL with *Fgf2* at 90 dps. Histological analysis and pixel classification was performed in the same way as in Fig. 4. The dotted boxes indicate the regions shown in the right panels. (H) Symmetry scores of each class. Scores obtained from the same limb are plotted at the same x-coordinates. The plots of “int” are the same plots as in Fig. 4. n.s.; no significant difference, * $p < 0.05$, ** $p < 0.005$. Scale bar = 700 μm (C), 4 mm (F, G, upper panels), 1 mm (F, G, lower panels).

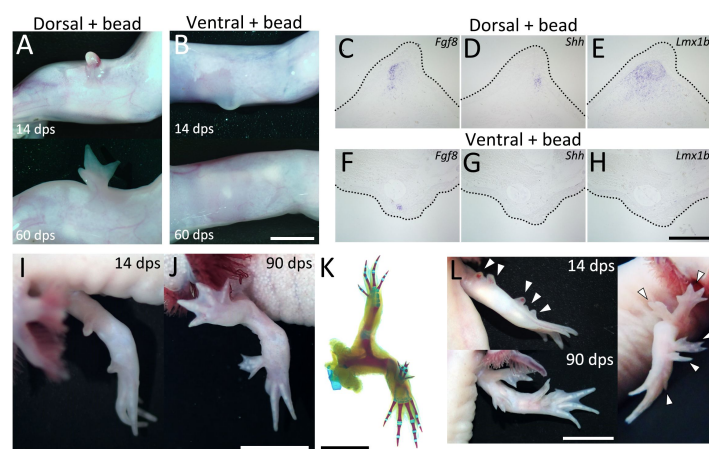


Figure 6.

Limb formation at the dorsal region by BMP2+FGF2+FGF8 supplementation.

(A, B) 14 and 60 dps phenotypes of BMP2+FGF2+FGF8 supplementation by bead grafting at the dorsal (A) or ventral (B) region. Neither nerve deviation nor skin grafting was performed. (C–H) Expression patterns of *Fgf8*, *Shh* and *Lmx1b* of induced blastemas at 10 dps. Expression was visualized by *in situ* hybridization. The dotted line indicates the external shape of the blastema. (I– K) Phenotype obtained by BMP2+FGF2+FGF8 supplementation to two dorsal regions of an identical limb. (L) Phenotype obtained by BMP2+FGF2+FGF8 supplementation to five dorsal regions of an identical limb. Scale bar = 3 mm (B), 700 μ m (H), 2 cm (J), 1 cm (K, L).

Discussion

Dorsal and ventral positional cues are required for the induction of *Shh* expression

In axolotl limb regeneration, cells derived from all four positional origins—anterior, posterior, dorsal, and ventral—are required for an induced blastema to form a limb. This was confirmed through the ALM experiments (Fig. 1). The gene expression patterns in these ALM blastemas revealed distinct molecular characteristics for each group (Fig. 2). *Fgf8* expression in AntBL and *Shh* expression in PostBL are consistent with a previous report (Nacu et al., 2016). The absence of SHH in AntBL and of FGF8 in PostBL likely accounts for their failure to form limbs. *Lmx1b* expression in the dorsal half of both AntBL and PostBL suggests that both dorsal and ventral tissues co-existed in these blastemas (Fig. 2B, G). In contrast, DorBL and VentBL exhibited distinct *Lmx1b* expression patterns, suggesting that most cells in DorBL and VentBL are derived from dorsal or ventral origins, respectively (Fig. 2L, Q). A possible explanation for the lack of *Shh* expression in DorBL and VentBL, despite the expression of *Fgf8* (Fig. 2M, N, R, S), is that the induction of *Shh* expression may depend on the co-existence of dorsal and ventral cells, although *Fgf8* can be expressed independently of the co-existence of dorsal and ventral cells. This idea is supported by the cell-tracing experiment (Fig. 3). In these assays, the posteriorly derived cells expressed *Shh* only when both dorsal and ventral cells were present. These results strongly suggest that the co-existence of dorsal and ventral cells is necessary for inducing *Shh* expression. Consequently, one of the essential roles of the dorsal and ventral positional cues in limb patterning would be inducing *Shh* expression, which facilitates the anteroposterior interaction.

Our findings clarify the hierarchical relationship between dorsal and ventral positional cues and anteroposterior interaction. Our data demonstrate that dorsal and ventral cells are essential for inducing *Shh* expression in a regenerating blastema. Furthermore, we showed that DorBL and VentBL, which typically fail to form a limb, could form patterned limbs with ectopic *Shh* expression, even in the absence of dorsal and ventral cell co-existence (Fig. 4). Furthermore, such limbs exhibited dorsally or ventrally symmetric structures (Fig. 4F, G, J–R), highlighting that dorsoventral patterning depends on the cell origin. Thus, we concluded that dorsal and ventral positional cues are essential for patterned limb formation via inducing *Shh* expression.

WNT10B and FGF2 as dorsal and ventral positional cue molecules

We identified *Wnt10b* and *Fgf2* as candidate genes providing the dorsal and ventral positional cues required to induce *Shh* expression, respectively (Fig. 5). DEG analysis between DorBL and VentBL revealed higher *Wnt10b* expression in DorBL and higher *Fgf2* expression in VentBL (Fig. 5A, B). Supplementation with WNT10B in VentBL and FGF2 in DorBL induced *Shh* expression and subsequent limb patterning (Fig. 5C–G). *Wnt4* expression was also elevated in DorBL, while *Fgf7* and *Tgfb2* were upregulated in VentBL. However, we did not observe *Shh* induction in the *Fgf7*- or *Tgfb2*-electroporated DorBL or in the *Wnt4*-electroporated VentBL. FGF7 is known to be capable of inducing the apical ectodermal ridge (AER) in chick limb development (Yonei-Tamura et al., 1999), but little is known about its role in limb regeneration. In newt limb regeneration, KGFR, which acts as FGF7 receptor, is expressed in the basal layer of the wound epithelium, while FGFR1, which acts as FGF2 receptor, is primarily expressed in the mesenchyme (Poulin et al., 1993). This differential expression pattern suggests that FGF2 and FGF7 target distinct cell populations, potentially explaining the differences in their ability to induce *Shh* expression. The difference in the ability of *Wnt10b* and *Wnt4* to induce *Shh* expression in VentBL may be explained by their distinct signaling pathways. WNT10B and WNT4 are typically classified as canonical and non-canonical WNTs, respectively. We observed *Shh* induction in BIO-treated VentBL, indicating that the canonical WNT signaling pathway regulates *Shh* expression. These results support the

idea that canonical, but not non-canonical, WNT signaling contributes to *Shh* induction. Regarding *Tgfb2*, TGF- β signaling has been implicated in blastema induction during salamander limb regeneration (Sader and Roy, 2022 [DOI](#); Lévesque et al., 2007 [DOI](#)). However, little is known about the relationship between TGF- β signaling and *Shh* induction in limb development and regeneration, suggesting that TGF- β signaling does not play a primary role in *Shh* induction. Consistent with this, we did not observe *Shh* induction in the *Tgfb2*-electroporated DorBL, as described above. This suggests that TGF β 2 signaling may be involved in dorsoventral regulation independent of *Shh* induction. Taken together, among the candidate genes we identified, WNT10B via the canonical WNT pathway and FGF2 via FGFR1 appear to regulate *Shh* induction and the subsequent patterning process.

WNT signaling may be the key for the dorsal properties in limb formation. It has been reported that canonical Wnt/ β -catenin signaling plays essential roles in limb regeneration among vertebrates, including axolotls (Kawakami et al., 2006 [DOI](#); Lovely et al., 2022 [DOI](#)). In the present study, we demonstrated that introducing *Wnt10b* or BIO treatment could induce *Shh* expression in VentBL, facilitating limb patterning. In mouse limb development, *Wnt10b* is expressed in the apical ectodermal ridge (AER, Witte et al., 2009 [DOI](#)), and mutations in *Wnt10b* are associated with Split-Hand/Foot Malformation (Ugur and Tolun, 2008 [DOI](#); Al Ghamdi et al., 2020 [DOI](#); Bilal et al., 2023 [DOI](#)).

However, there is no evidence suggesting that *Wnt10b* contributes to dorsal specificity during limb development. This raises questions about whether the function of *Wnt10b* is unique to axolotls or whether its orthologs in other species might share functional redundancy with other WNT genes. Further studies are needed to clarify this point. Notably, WNT10B utilizes the canonical Wnt signaling pathway, similar to WNT7A, a well-known WNT family gene critical for dorsoventral limb patterning in amniotes. In amniotes, *Wnt7a* is expressed in the dorsal ectoderm of developing limb buds and induces *Lmx1b* expression in dorsal mesenchyme, thereby acting as positional information to establish dorsal characteristics (Riddle et al., 1995 [DOI](#); Cygan et al., 1997 [DOI](#); Chen, 2002 [DOI](#)). In axolotls, dorsal-specific *Wnt7a* expression has not been confirmed. Our RNA-seq analysis showed that there was almost no significant difference in *Wnt7a* expression levels between DorBL and VentBL ($\log_2$ (normalized counts) = 5.97, $\log_2$ (fold change) = -0.810, $p = 0.401$), consistent with previous studies (Shimokawa et al., 2013 [DOI](#)). These results suggest that *Wnt7a* does not have a dorsal-specific function, at least in axolotl limb regeneration/development. While WNT10B could function as a dorsal positional cue, WNT10B is unlikely to induce dorsal identity, as ectopic *Lmx1b* expression was not observed in *Wnt10b*-introduced VentBL, which formed double-ventral limbs (Fig. 5F–H [DOI](#)). This indicates that *Wnt10b* in axolotl limb regeneration does not simply replace the function of *Wnt7a* in amniote limb development. Nevertheless, the involvement of canonical WNT signaling in dorsal function and limb morphogenesis remains an important area for further investigation.

We identified FGF2 as the ventral positional cue, which plays a crucial role in limb patterning process in axolotl limb regeneration. This aligns with previous studies showing that *Fgf2* expression correlates with limb regeneration in salamanders (Giampaoli et al., 2003 [DOI](#)). *Fgf2* expression was relatively high in VentBL, and *Fgf2*-introduced DorBL formed patterned limbs (Fig. 5C [DOI](#)). These results indicate that FGF2 supplementation can substitute for the presence of ventral cells in the limb patterning process. It is noteworthy that FGF2 application does not appear to induce ventral identity, as the limbs formed from *Fgf2*-introduced DorBL exhibited dorsally symmetric limb structures. The finding that FGF signaling functions in dorsoventral control is not a well-known mechanism across species studied to date. Whether this represents an axolotl-specific regulatory mechanism or a broader function of FGF signaling in limb morphogenesis remains to be determined.

Our findings suggest that although WNT10B and FGF2 act as dorsal and ventral positional cues, they do not affect dorsal or ventral identities. In amniote limb development, WNT7A functions as dorsal “positional information” by inducing dorsal identity through *Lmx1b* expression. In contrast, in axolotls, supplementation with WNT10B in VentBL or FGF2 in DorBL did not affect the expression patterns of *Lmx1b* (Fig. 5C). Moreover, the limbs formed from such DorBL and VentBL exhibited dorsally and ventrally symmetric structures (Fig. 5F–H). These findings suggest that dorsoventral identities in axolotls are not affected by WNT10B or FGF2 supplementation. Thus, in the present study, we used the term “positional cue” to distinguish this role from “positional information,” which provides positional identities. We previously reported that the expression patterns of *Lmx1b* in axolotl limb regeneration are likely to depend on the positional origins of cells (Iwata et al., 2020; Yamamoto et al., 2022). The identities of cells along the dorsoventral axis may be controlled by their positional memory and determined before the initiation of limb regeneration.

The present study revealed that the dorsal and ventral positional cue molecules WNT10B and FGF2 regulate *Shh* expression during limb patterning in axolotls. These findings highlight a degree of conservation between axolotl limb regeneration and amniote limb development. In amniote limb development, FGF and WNT signaling pathways are key regulators of *Shh* expression. In developing amniote limb buds, *Fgf2* is expressed in the ectoderm, including AER and adjacent mesoderm, mediating distal outgrowth. FGFs, including FGF2, from AER are known to maintain *Shh* expression (Laufer et al., 1994; Niswander et al., 1994; Yang and Niswander, 1995; Li et al., 1996). Similarly, WNT7A regulates *Shh* expression in amniote limb development, and loss of WNT7A function results in reduced *Shh* expression in the zone of polarizing activity (ZPA) and the deletion of posterior structures (Parr and McMahon, 1995). Thus, our findings provide new insights into the conserved and divergent roles of dorsal-ventral signaling in regulating *Shh* expression, furthering our understanding of limb morphogenesis.

The integration mechanism of the four positional cues

In the present study, we reveal the molecular hierarchy of the four positional cues— anterior, posterior, dorsal, and ventral—in axolotl limb regeneration and explain how these positional cues are integrated to form a patterned limb (Fig. 7). In our proposed model, following amputation, nerves first trigger blastema induction by secreting nerve factors, such as BMP and FGFs (Makanae et al., 2014; Satoh et al., 2016). These nerve factors stimulate connective tissue cells, including dermal cells, to generate multipotent mesenchymal blastemal cells (Muneoka et al., 1986; Kragl et al., 2009; Hirata et al., 2010; Gerber et al., 2018). Simultaneously, *Wnt10b* and *Fgf2* are expressed at higher levels in dorsal and the ventral blastemal cells, respectively. In the next phase, the co-existence of WNT10B and FGF2 induces *Shh* expression in the posterior region of the blastema. During this phase, *Fgf8* is expressed in the anterior mesenchyme independently of the co-existence of dorsal and ventral cells (Fig. 7A). These expression patterns of *Wnt10b*, *Fgf2*, *Shh*, and *Fgf8* may be determined by the positional memory of cells (Otsuki et al., 2021). In the subsequent phase, the anteroposterior interaction, mediated by FGF8 and SHH, supports distal outgrowth to form a limb. This model explains previous observations in studies on double-half limbs and ALM blastemas (AntBL, PostBL, DorBL, and VentBL), which typically fail to form a limb. In blastemas induced from double-anterior and double-posterior limbs, or AntBL (Fig. 7B) and PostBL (Fig. 7C), SHH or FGF8 proteins are likely absent due to the lack of anteriorly or posteriorly derived cells. Similarly, in blastemas induced by amputating double-dorsal and double-ventral limbs, or DorBL (Fig. 7D) and VentBL (Fig. 7E), FGF2 or WNT10B proteins are likely absent or insufficient because of the lack of ventrally or dorsally derived cells. This results in the absence of SHH, even if posteriorly derived cells are present. In all these cases, the absence of either FGF8 or SHH disrupts the anteroposterior interaction, leading to failure in limb patterning. We concluded that the requirement for cells derived from all four positional origins is underpinned by this integration model. In this model, dorsal and ventral positional cues activate the posterior positional cue, enabling mutual interaction with the anterior cue. This interplay ensures proper anteroposterior interaction and complete limb patterning.

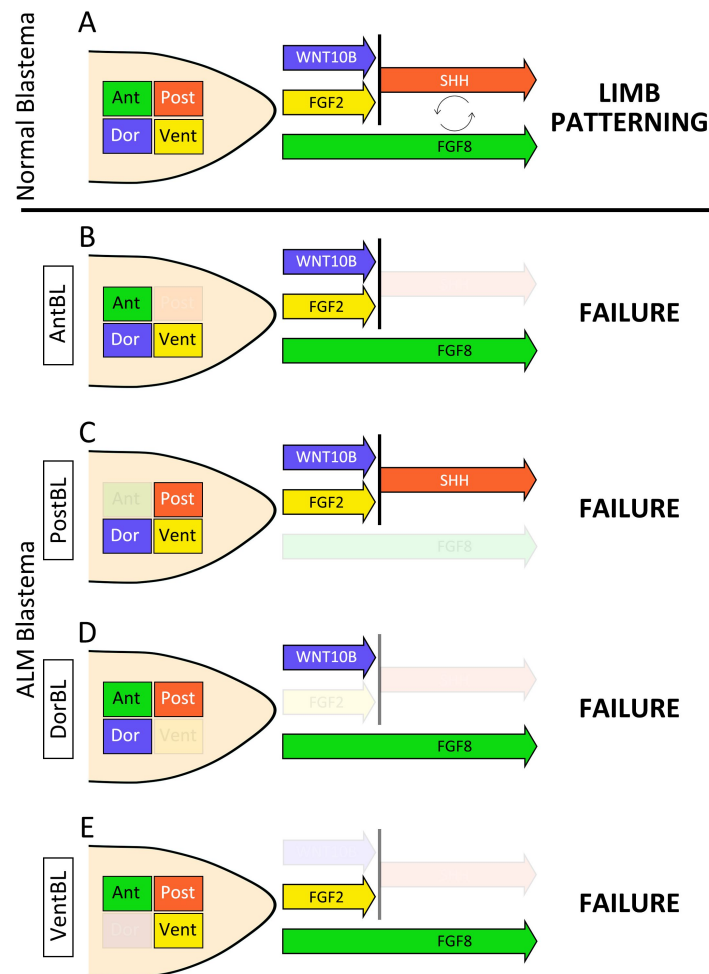


Figure 7.

The integration model of the four positional cues.

Schematic images of a normal blastema (A), AntBL (B), PostBL (C), DorBL (D), and VentBL (E). Green, red, blue, and yellow boxes within the blastema represent cells derived from anterior, posterior, dorsal, and ventral regions, respectively. Colored arrows indicate the presence of the corresponding positional cue molecules. In this model, limb patterning requires both FGF8 and SHH, and *Shh* expression in posteriorly derived cells is induced by the co-existence of WNT10B and FGF2.

Materials and Methods

Animal procedures

Axolotls between 4 and 15 cm from snout to tail tip were housed in tap water at 22°C. Both forelimbs and hind limbs were used for surgical procedures without distinction. Transgenic axolotls were obtained from the Ambystoma Genetic Stock Center (<http://www.ambystoma.org/genetic-stock-center>). Animals were anesthetized in 0.1% MS222 (Sigma-Aldrich) solution before surgical procedures.

In the ALM experiments, skin was peeled from the anterior, posterior, dorsal, or ventral side of a limb at the stylopod level, so that the skin of the opposite side was not injured. Thus, the size of the injured area depended on the limb size. Then, thick bundles of nerve trunks running the center of a limb was deviated the injured region. For the cell-tracing experiment, a piece of posterior half of the dorsal or ventral skin was obtained from GFP-expressing transgenic animals and grafted to the ALM region.

Protein-soaked beads for grafting were prepared as previously described (Makanae et al., 2014; Kashimoto et al., 2023). In brief, air-dried gelatin beads were allowed to swell in stock solution (1 µg/µL) prepared following the manufacturer's instructions. Equal amounts of proteins were used when formulating the combination protein mixture. Beads were soaked in the mixture of proteins (Bmp2 [mouse], Fgf2 [mouse], and Fgf8 [human/mouse]; R&D Systems) overnight at 4°C. For control experiments, gelatin beads were soaked in DDW. Before grafting, dorsal or ventral skin was peeled because full-thickness skin inhibits limb regeneration (Thornton, 1962; Mescher, 1976; Tsai, 2020). The beads were grafted under the wounded epidermis at 3 dps. The limbs were fixed 10 days after grafting. The details of grafting procedures are as previously described (Makanae et al., 2014; Kashimoto et al., 2023).

To analyze skeletal patterns, induced limbs were stained with alcian blue and alizarin red. Samples were fixed in 100% ethanol for 1 day at room temperature, then stained with Alcian blue solution (Wako, pH 2.0) in 20% acetic acid (Nacalai Tesque) with 80% ethanol solution at 37°C overnight. Then, samples were washed in tap water several times and fixed in 10% Formaldehyde Neutral Buffer Solution (Nacalai Tesque) for 1 day. Samples were then stained with Alizarin red S (Nacalai Tesque) in the solution (4% KOH:10% Formaldehyde Neutral Buffer Solution = 2:3) at room temperature for 1 day. Finally, samples were placed in graded glycerol with 4% KOH for clearing.

For BIO treatment, 10 mM stock solution of 6-bromoindirubin-3-oxime (BIO, Selleck, S7198) dissolved in DMSO was stored in the dark at 4°C. Axolotls soon after surgery were raised in tap water with BIO solution (experimental) or with the same amount of solvent, DMSO (control), until 14 dps for qPCR and 30 dps for phenotype analysis. The containers including water and axolotls were kept in the dark during BIO or DMSO treatment. For BIO treatment, axolotls between 4 and 5 cm from snout to tail tip were used.

Sectioning and histological staining

Samples were fixed in 4% PFA/PBS overnight at room temperature before sectioning. The fixed samples were embedded in O.C.T. compound (Sakura) following 30% sucrose/PBS treatment for approximately 12 h at 4°C. Frozen sections of 14 µm thickness were prepared using Leica CM1850 (Nussloch). The sections were well dried under an air dryer and kept at -80°C until use.

Standard hematoxylin and eosin (HE) staining and Masson's trichrome staining were used for histological analysis. To visualize cartilage formation, Alcian blue staining was performed before HE staining. In brief, sections were washed in tap water several times to remove the O.C.T. compound. Then, the sections were stained with Alcian blue (Wako, pH 2.0), and then HE staining was performed. Trichrome stain (Masson) kit (Sigma, HT15-1KT) was used for trichrome staining. The stained sections were mounted using Softmount (Wako).

In situ hybridization

In situ hybridization was performed as described previously (Yamamoto et al., 2022 [DOI](#)). For probe synthesis, target genes cloned on pTAC-2 plasmid (BioDynamics Lab Inc.) were amplified by PCR using M13 primers. PCR fragments were purified and used as an RNA probe template. RNA probe synthesis was performed with Sp6 RNA polymerase (Takara) for 3 h, and RNA was hydrolyzed for 11 min. The sections were washed in PBT to remove the O.C.T. compound, treated with proteinase K (10 µg/mL) (Invitrogen)/PBT for 20 min at room temperature, washed in PBT, treated with 4% PFA/PBS for 20 min at room temperature, washed in PBT, and then probes were hybridized at 62.5°C for approximately 18 h. The sections were washed in wash buffer 1 (formamide:H₂O:20× SSC [3M NaCl:0.3M sodium citrate, pH 5.0] = 2:1:1), and then in wash buffer 2 (formamide:H₂O:20× SSC = 5:1:4). The samples were then incubated with anti-digoxigenin-AP Fab fragments (Sigma-Aldrich, 1/1000) for 2 h at room temperature. Samples were stained with BCIP (Nacalai Tesque) and NBT (Nacalai Tesque) in alkaline phosphatase buffer (0.1M NaCl, 0.1M Tris-HCl [pH 9.5], 0.1% Tween20) for 18 h at room temperature after washing in TBST.

Immunofluorescence

Immunofluorescence on sections was carried out based on a previous report (Yamamoto et al., 2022 [DOI](#)). The antibodies were as follows: anti-GFP (MBL, #598, 1/500), anti-acetylated alpha tubulin (Santa Cruz, #sc-23950, 1/1000), anti-mouse IgG Alexa 488 (Invitrogen, #A11017, 1/1000), anti-rabbit IgG Alexa 488 (Invitrogen, #A21206, 1/500). Nuclei were stained with Hoechst (Nacalai Tesque), and images were captured using an Olympus BX51 system.

RNA-Seq analysis

Total RNA was extracted from DorBL and VentBL at 10 dps using TriPure reagent (Roche), following the manufacturer's instructions. Three biological replicates were prepared for both samples. 150 bp paired-end RNA-seq reads were obtained under contract to Rhelixa (Tokyo, Japan), using Illumina Nova Seq 6000 and SMART-Seq HT Plus Kit (#R400449). RNA-seq data were analyzed on Galaxy (<https://usegalaxy.eu/root> [DOI](#)) as follows. Sequence reads were trimmed and the quality was filtered by Trimmomatic v0.39 (Bolger et al., 2014 [DOI](#)) with the following parameters (LEADING:20, TRAILING:20, SLIDINGWINDOW:4:15, MINLEN:36). Axolotl genome data (AmexG_v6.0-DD) and annotation data (AmexT_v47-AmexG_v6.0-DD.gtf) were obtained from AXOLOTL-OMICS (<https://www.axolotl-omics.org/> [DOI](#)) (Schloissnig et al., 2021 [DOI](#)). Mapping to the Axolotl genome (AmexG_v6.0-DD) was performed with HISAT2 v2.2.1 (Kim et al., 2015 [DOI](#)). Count data were obtained with featureCounts v2.0.8 (Liao et al., 2013 [DOI](#)) on the basis of the Axolotl gene model (AmexT_v47-AmexG_v6.0-DD.gtf). DEGs were identified with DESeq2 v2.11.40.8 (Love et al., 2014 [DOI](#)) ($p < 0.05$). Among identified DEGs (DorBL > VentBL; 762 genes, VentBL > DorBL; 513 genes), genes annotated as “intercellular signaling molecule” were explored on PANTHER (<https://www.pantherdb.org/> [DOI](#)) (Thomas et al., 2003 [DOI](#)), and 21 genes (DorBL > VentBL; 5 genes, VentBL > DorBL; 16 genes) were identified as candidate genes of dorsal and ventral positional cues, as shown below:

DorBL > VentBL:

WNT4 (AMEX60DD052091), WNT10B (AMEX60DD029981), SEMA3A (AMEX60DD023165), LECT2 (AMEX60DD041044), ANGPTL5 (AMEX60DD049512).

VentBL > DorBL:

FCN1 (AMEX60DD050926), CXCL14 (AMEX60DD028973), DNER (AMEX60DD002010), FGF7 (AMEX60DD003767), CXCL12 (AMEX60DD052412), CXCL8 (AMEX60DD044133), SEMA6A (AMEX60DD042813), ANGPTL2 (AMEX60DD050490), ANGPTL6 (AMEX60DD031854), FGL2 (AMEX60DD006387), VWDE (AMEX60DD022491), TGFB2 (AMEX60DD036126), FGF2 (AMEX60DD044865), EFNA4 (AMEX60DD014882), EFNA3 (AMEX60DD014867), ANGPTL1 (AMEX60DD018261).

Cloning candidate genes and preparation of construct vectors for overexpression

Among the identified candidate genes, we focused on *Wnt10b*, *Wnt4*, *Fgf2*, *Fgf7*, and *Tgfb2*. We cloned these genes in pTAC-2 plasmids from cDNA obtained from blastemas. The following primers were used:

Wnt10b Fwd:ATGGCCACAGCTCACCTCCGACACC

Wnt10b Rev:TCACTTGACACATTCACCCATTCTGTG

Wnt4 Fwd:ATGGATGCTACGAAAGCAGCGTATATC

Wnt4 Rev:TCACCGGCAGGTGTGCATTCTACCAC

Fgf2 Fwd:ATGGCGGCGGGGAGCATCACCACTTGC

Fgf2 Rev:TCAACTCTTGCCGACATGGGAAGGAAAAG

Fgf7 Fwd:ATGCGCAGATGGGTGCTAGCTTGGATC

Fgf7 Rev:TCATGTGTTATTGGATATACGCATTGGA

TGFB2 Fwd:ATGAGATTACAATTACTGAGAAAAAAAAAATG

TGFB2 Rev:TTAGCTGCACTTGCAAGATTTTACAATCA

We then inserted these genes to p2a-AcGFP-pCS2 vectors with In-Fusion® HD Cloning Kit (Clontech). The following primers were used for PCR before In-Fusion reaction:

pCS2 inverse Fwd:GCTACTAACTTCAGCCTGCTGAAGCAGG

pCS2 inverse Rev:CATCGATGGGATCCTGCAAAAAGAACAAGTAGCTT

Wnt10b Fwd:AGGATCCCATCGATGGCCACAGCTCACCTCCGA

Wnt10b Rev:GCTGAAGTTAGTAGCCTTGACACATTCACCCA

Wnt4 Fwd:AGGATCCCATCGATGGATGCTCACGAAAGCAGCG

Wnt4 Rev:GCTGAAGTTAGTAGCCCGGCAGGTGTGCATTCTA

Fgf2 Fwd:AGGATCCCATCGATGGCGGCGGGGAGCATACCA

Fgf2 Rev:GCTGAAGTTAGTAGCACTCTTGCCGACATGGGAA

Fgf7 Fwd:AGGATCCCATCGATGCGCAGATGGGTGCTAGCTTG

Fgf7 Rev:GCTGAAGTTAGTAGCTGTGTTATTGGATATACGCA

Tgfb2 Fwd:AGGATCCCATCGATGAGATTACAATTACTGAGAAA

Tgfb2 Rev:GCTGAAGTTAGTAGCGCTGCACTTGCAAGATTTTA

Finally, the following DNA constructs were obtained; *Wnt10b*-p2a-GFP (pCS2), *Wnt4*-p2a-GFP (pCS2), *Fgf2*-p2a-GFP (pCS2), *Fgf7*-p2a-GFP (pCS2), and *Tgfb2*-p2a-GFP (pCS2).

Electroporation

Each DNA construct was injected directly into the target region. Immediately after injection, electric pulses were applied (20 V, 50 ms pulse length, 950 ms interval, 10 times) with NEPA21(Nepa gene). The injected DNA constructs were as follows: pCS2-AcGFP, pCS2-*Shh*-p2a-AcGFP, *Wnt10b*-p2a-GFP (pCS2), *Wnt4*-p2a-GFP (pCS2), *Fgf2*-p2a-GFP (pCS2), *Fgf7*-p2a-GFP (pCS2), and *Tgfb2*-p2a-GFP (pCS2). All plasmids were purified using a Genopure Maxi kit (Roche). Electroporation was performed at -3, 4, and 7 dps and samples were fixed at 10 dps for *in situ* hybridization. To analyze the phenotype of 90 dps samples, electroporation was performed at -3, 4, 7, 10, 15, 20, 25, and 30 dps.

qRT-PCR

The procedures of qRT-PCR were previously described (Yamamoto et al., 2022 [\[1\]](#)). In brief, RT was performed using Prime Script II (Takara), and qPCR was performed using KAPA SYBR FAST qPCR Master Mix (Kapa Biosystems) and StepOne (ThermoFisher Scientific). Primers were as follows:

Ef-1α Fwd: AACATCGTGGTCATCGGCCAT

Ef-1α Rev: GGAGGTGCCAGTGATCATGTT

Shh Fwd: GCTCTGTGAAAGCAGAGAACTCG

Shh Rev: CGCTCCGTCTCTATCAGTAGAA

RNA for **Fig. 5D** [\[1\]](#) was prepared from FGF2- or DDW-soaked bead grafted DorBL. Bead grafting was performed at 7 dps and RNA preparation was performed at 14 dps. RNA for **Fig. 5E** [\[1\]](#) was prepared from 1μM BIO- or DMSO-treated VentBL at 14 dps. *Ef-1α* was used as the internal control. We conducted a two-tailed Welch's *t*-test for statistical analysis.

Pixel classification and calculating symmetry scores

A machine learning-based method was applied for pixel classification using ilastik software (downloaded on <https://www.ilastik.org/> [\[1\]](#)). The details and workflows were previously described (Berg et al., 2019 [\[1\]](#)). Each pixel in the images was classified into five classes. The regions of the background (Class 1), cartilage (Class 2), muscle (Class 3), other connective tissue (Class 4), and epidermis (Class 5) were annotated for training data. Using these annotated regions as references, pixels were automatically classified into the respective classes. In this process, the same training data were used for images obtained from the same limb. Then, symmetry scores were calculated for each class individually and for the combined set of all classes. In this process, the external shape of the section, which could bend randomly during sample fixation process, would affect the symmetry scores if the entire region were used. To focus on the symmetry of the anatomical patterns, images with 400 μm width were prepared (**Fig. 4L** [\[1\]](#), **5E** [\[1\]](#), F). The symmetry scores of pixel-classified images were calculated using python. First, the center of the dorsal end and ventral

end was set as the axis of symmetry. Then, one side of the image was flipped. Next, color masks were generated for both sides of the image. These masks identified pixels that matched the specified color. Pixels were considered to match if their RGB values were within the given tolerance range for all three channels. In pixel comparison, the masks for one side and the other side were compared to calculate the following pixels:

Matching pixels: The number of pixels that match in both sides for the specified color. Total pixels: The total number of pixels matching the color in either side.

The symmetry scores were computed as (Number of Matching Pixels)/(Total Pixels in Both Sides). The scores were obtained from 12 areas of each group. For statistical analysis, a two-tailed a two-tailed Welch's *t*-test was used. In this analysis, each group was compared to the intact limb group because the intact limb should be set as a typical asymmetrical structure.

Additional information

Author contributions

Conceptualization: S.Y., A.S.; Methodology: S.Y., M.H., A.S.; Validation: S.Y.; Formal analysis: S.Y.; Investigation: S.Y.; Resources: S.Y., S.F., A.O., A.S.; Data curation: S.Y.; Writing - original draft: S.Y.; Writing - review & editing: S.Y., S.F., A.O., M.H., A.S.; Visualization: S.Y., A.S.; Supervision: A.S.; Project administration: A.S.; Funding acquisition: A.S., S.Y.

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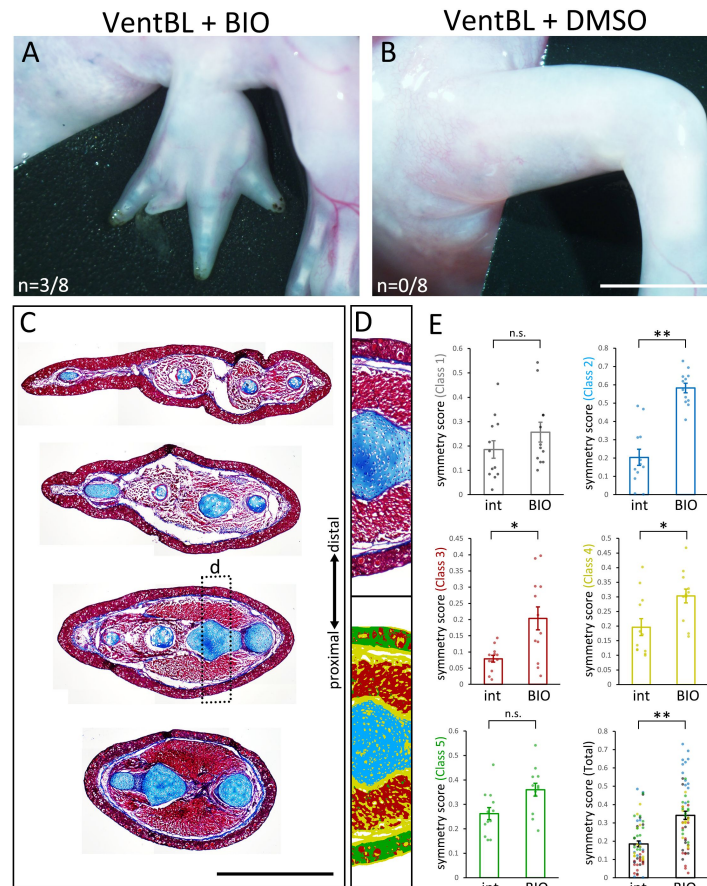


Figure S1.

Limb patterning from BIO-treated VentBL.

(A, B) VentBL treated with 1 μ M BIO (A) or DMSO (B) at 90 dps. (C–D) Histological analysis and pixel classification was performed in the same way as in Fig. 4. The dotted boxes indicate the regions shown in (D). (E) Symmetry scores of each class. Scores obtained from the same limb are plotted at the same x-coordinates. The plots of “int” are the same plots as in Fig. 4. n.s.; no significant difference, * p <0.05, ** p <0.005. Scale bar = 4 mm (B), 1 cm (C).

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Reviewer #1 (Public review):

Summary:

The manuscript by Yamamoto et al. presents a model by which the four main axes of the limb are required for limb regeneration to occur in the axolotl. A longstanding question in regeneration biology is how existing positional information is used to regenerate the correct missing elements. The limb provides an accessible experimental system by which to study the involvement of the anteroposterior, dorsoventral, and proximodistal axes in the regenerating limb. Extensive experimentation has been performed in this area using grafting experiments. Yamamoto et al. use the accessory limb model and some molecular tools to address this question. There are some interesting observations in the study. In particular, one strength is the potent induction of accessory limbs in the dorsal axis with BMP2+Fgf2+Fgf8, which is very interesting.

Strengths:

The manuscript presents some novel phenotypes generated in axolotl limbs due to Wnt signaling. This is generally the first example in which Wnt signaling has provided a gain-of-function in the axolotl limb model. They also present a potent way of inducing limb patterning in the dorsal axis by the addition of just beads loaded with Bmp2+Fgf8+Fgf2.

Weaknesses:

Although interesting, the study makes bold claims about determining the molecular basis of DV positional cues, but the experimental evidence is not definitive and does not take into account the previous work on DV patterning in the amniote limb. Also, testing the hypothesis on blastemas after limb amputation would be needed to support the strong claims in the study. There are several examples of very strong claims, but the evidence lacks support for these claims.

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Reviewer #2 (Public review):**Summary:**

This study explores how signals from all sides of a developing limb, front/back and top/bottom, work together to guide the regrowth of a fully patterned limb in axolotls, a type of salamander known for its impressive ability to regenerate limbs. Using a model called the Accessory Limb Model (ALM), the researchers created early limb regenerates (called blastemas) with cells from different sides of the limb. They discovered that successful limb regrowth only happens when the blastema contains cells from both the top (dorsal) and bottom (ventral) of the limb. They also found that a key gene involved in front/back limb patterning, called *Shh* (Sonic hedgehog), is only turned on when cells from both the dorsal and ventral sides come into contact. The study identified two important molecules, *Wnt10B* and *FGF2*, that help activate *Shh* when dorsal and ventral cells interact. Finally, the authors propose a new model that explains how cells from all four sides of a limb, dorsal, ventral, anterior (front), and posterior (back), contribute at both the cellular and molecular level to rebuilding a properly structured limb during regeneration.

Strengths:

The techniques used in this study, like delicate surgeries, tissue grafting, and implanting tiny beads soaked with growth factors, are extremely difficult, and only a few research groups in the world can do them successfully. These methods are essential for answering important questions about how animals like axolotls regenerate limbs with the correct structure and orientation. To understand how cells from different sides of the limb communicate during regeneration, the researchers used a technique called *in situ* hybridization, which lets them see where specific genes are active in the developing limb. They clearly showed that the gene *Shh*, which helps pattern the front and back of the limb, only turns on when cells from both the top (dorsal) and bottom (ventral) sides are present and interacting. The team also took a broad, unbiased approach to figure out which signaling molecules are unique to dorsal and ventral limb cells. They tested these molecules individually and discovered which could substitute for actual dorsal and ventral cells, providing the same necessary signals for proper limb development. Overall, this study makes a major contribution to our understanding of how complex signals guide limb regeneration, showing how different regions of the limb work together at both the cellular and molecular levels to rebuild a fully patterned structure.

Weaknesses:

Because the expressional analyses are performed on thin sections of regenerating tissue, they provide only a limited view of the gene expression patterns in their experiments, opening the possibility that they could be missing some expression in other regions of the blastema. Additionally, the quantification method of the expressional phenotypes in most of the experiments does not appear to be based on a rigorous methodology. Therefore, performing alternate expressional analysis, using RNA-seq or qRT-PCR (for example) on the entire blastema would help validate that the authors are not missing something.

Overall, the number of replicates per sample group is quite low (sometimes as low as 3), which is especially risky with challenging techniques like the ones the authors employ. The authors don't appear to have performed a power analysis to calculate the number of animals used in each experiment that are sufficient to identify possible statistical differences between groups. Increasing the sample sizes would substantially increase the rigor of their experiments.

Likewise, the authors' use of an AI-generated algorithm to quantify symmetry on the dorsal/ventral axis, and this approach doesn't appear to account for possible biases due to tissue sectioning angles. They also appear to arbitrarily pick locations in each sample group

to compare symmetry measurements. There are other methods, which include using specific muscle groups and nerve bundles as dorsal/ventral landmarks, that would more clearly show differences in symmetry.

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Reviewer #3 (Public review):

Summary:

After salamander limb amputation, the cross-section of the stump has two major axes: anterior-posterior and dorsal-ventral. Cells from all axial positions (anterior, posterior, dorsal, ventral) are necessary for regeneration, yet the molecular basis for this requirement has remained unknown. To address this gap, Yamamoto et al. took advantage of the ALM assay, in which defined positional identities can be combined on demand and their effects assessed through the outgrowth of an ectopic limb. They propose a compelling model in which dorsal and ventral cells communicate by secreting Wnt10b and Fgf2 ligands, respectively, with this interaction inducing Shh expression in posterior cells. Shh was previously shown to induce limb outgrowth in collaboration with anterior Fgf8 (PMID: 27120163). Thus, this study completes a concept in which four secreted signals from four axial positions interact for limb patterning. Notably, this work firmly places dorsal-ventral interactions upstream of anterior-posterior, which is striking for a field that has been focussed on anterior-posterior communication. The ligands identified (Wnt10b, Fgf2) are different from those implicated in dorsal-ventral patterning in the non-regenerative mouse and chick models. The results in the context of ALM/ectopic limb engineering are impressive, but the authors do not extend their experiments to assay 'normal' regeneration after amputation.

Strengths:

(1) The ALM and use of GFP grafts for lineage tracing (Figures 1-3) take full advantage of the salamander model's unique ability to outgrow patterned limbs under defined conditions. As far as I am aware, the ALM has not been combined with precise grafts that assay 2 axial positions at once, as performed in Figure 3. The number of ALMs performed in this study deserves special mention, considering the challenging surgery involved.

(2) The authors identify that posterior Shh is not expressed unless both dorsal and ventral cells are present. This echoes previous work in mouse limb development models (AER/ectoderm-mesoderm interaction), but this link between axes was not known in salamanders. The authors elegantly reconstitute dorsal-ventral communication by grafting, finding that this is sufficient to trigger Shh expression (Figure 3 - although see also the Weaknesses section.)

(3) Impressively, the authors discovered two molecules sufficient to substitute dorsal or ventral cells through electroporation into dorsal- or ventral-depleted ALMs (Figure 5). These molecules did not change the positional identity of target cells. The same group previously identified the ventral factor (Fgf2) to be a nerve-derived factor essential for regeneration. In Figure 6, the authors demonstrate that nerve-derived factors, including Fgf2, are alone sufficient to grow out ectopic limbs from a dorsal wound. Limb induction with a 3-factor cocktail without supplementing with other cells is conceptually important for regenerative engineering.

(4) The writing style and presentation of results are very clear.

Weaknesses:

(1) The expression data are the weakest part of this study.

- Despite being a central message, I found the Shh in situ unconvincing (e.g. Figure 2I, 3C, 5C), especially without sense probe controls. An additional assay would be essential to make the Shh data convincing - perhaps like in Figure 5D (qPCR?), RNA-sequencing, or a downstream target gene.
- It is not clear what the n numbers mean for the in situ data (slides analysed / number of biological samples / other?). This is crucial to understanding the reliability of the results.
- The authors do not assay where and when Wnt10b and Fgf2 are expressed beyond the bulk RNA-sequencing (which presumably contains both epidermis and mesenchyme cells). This is a shame, as understanding which cell types express these molecules, and when, would be important for understanding the mechanism.

(2) It is important to consider that the ALM is not 'regeneration', even if the authors have previously argued that ALM bumps and regenerating blastemas are equivalent (PMID: 17959163). The start- and end- points of ALM are different from regeneration, even though there are undoubtedly common principles involved. Thus, I find the word 'regeneration' in the title and last sentence of the abstract unsubstantiated unless evidence is provided that the same mechanisms (Wnt10b/Fgf2/Shh) function during normal limb regeneration.

(3) Drawing the exact boundaries of the Ant/Pos/Dor/Ven BL and grafts in the cartoon in Figure 1 (with respect to anatomical landmarks) would help to better understand the experiments in Figures 3 and 4.

(4) I find the 'positional cue' and 'positional value' terminology confusing, despite the authors' efforts. It is not clear if they refer to cell autonomous or secreted signals, and, as the authors mention, the definitions partially overlap. Lmx1b is defined as a positional value, even though it is necessary and sufficient for dorsal identity (so, isn't it positional information?). Much simpler would be to describe Wnt10b and Fgf2 as what they are: dorsally or ventrally expressed signals that substitute for dorsal or ventral tissue without inducing changes in positional information.

Overall appraisal:

This is a logical and well-executed study that creatively uses the axolotl model to advance an important framework for understanding limb patterning. The reliability of the Shh expression data is a weak point in this otherwise impressive study. The relevance of the mechanisms to normal limb regeneration is not substantiated.

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Author response:

We sincerely thank the editor and all three reviewers for their constructive comments. We deeply appreciate the reviewers' efforts in highlighting both the strengths and the weaknesses of our study. To enhance the quality and clarity of our work, we plan to address the concerns raised in the public reviews through the following actions:

(1) Improving the tone and language of the manuscript

We will revise the manuscript thoroughly, incorporating additional explanations and clarifications where necessary, and improving the tone and language to enhance readability and precision. Especially, we will pay careful attention on the terms “positional information,” “positional value,” and “positional cue,” and we plan to explain them in a historical context.

(2) Extending analysis to regular blastemas

To validate the applicability of our proposed model beyond the accessory limb model (ALM), we will examine the gene expression patterns of key signaling molecules in regular blastemas generated by limb amputation. This will allow us to test whether the mechanisms we describe are also active during normal limb regeneration.

(3) Increasing sample sizes in critical experiments

In order to ensure reproducibility and statistical reliability, we will increase the number of biological replicates in key experiments within the limitations regulated by our animal ethics approval. Additionally, we will collect data that clearly defines the dorsal/ventral axis within the structures, as far as possible. We will also revise the manuscript to pay closer attention to the anterior/posterior/dorsal/ventral axis in the existing data, ensuring that it is clearly described.

(4) Adding quantitative gene expression data

To support and reinforce our *in situ* hybridization results, we will include additional quantitative gene expression analyses (e.g., qRT-PCR), thereby strengthening the conclusions drawn from our expression data.

We are grateful for the reviewers' insights and are confident that these revisions will significantly strengthen our manuscript.

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